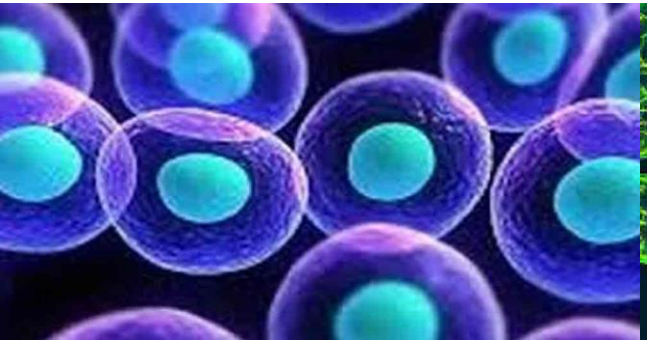
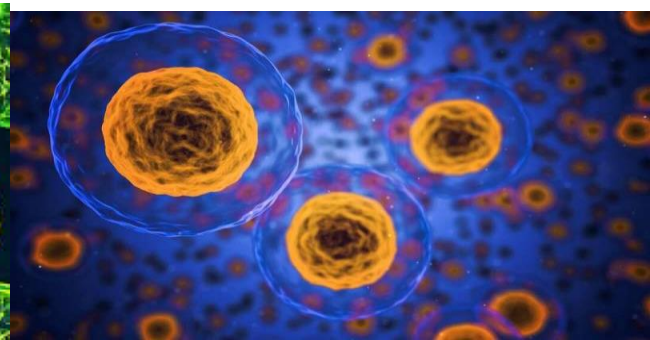
Part 5

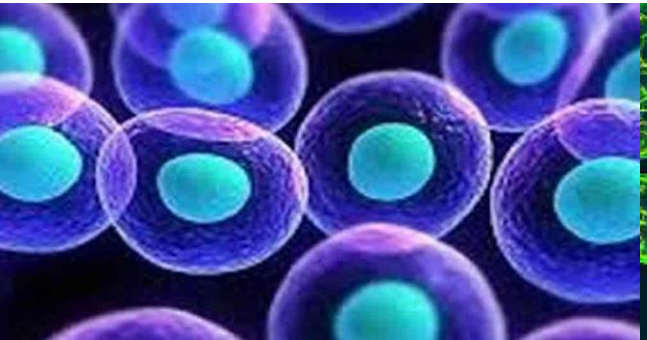
The nucleus

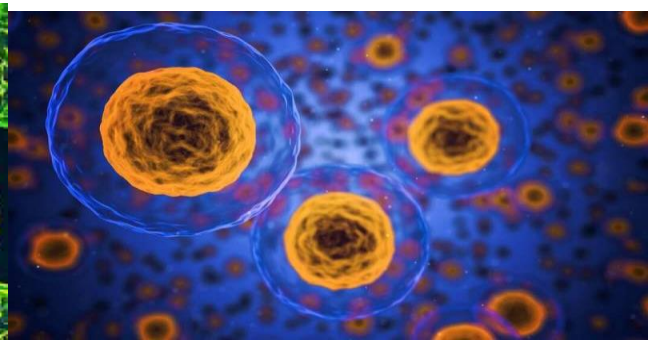


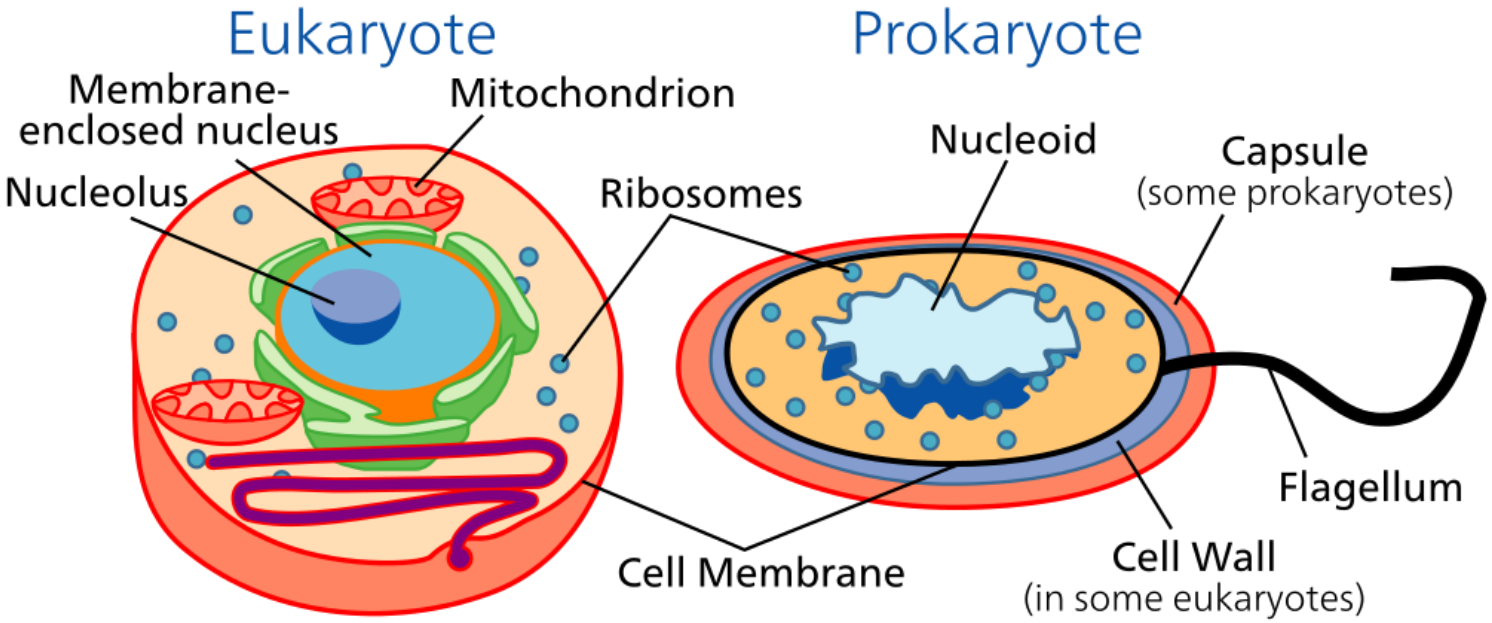


5.1

The nucleus - Structure







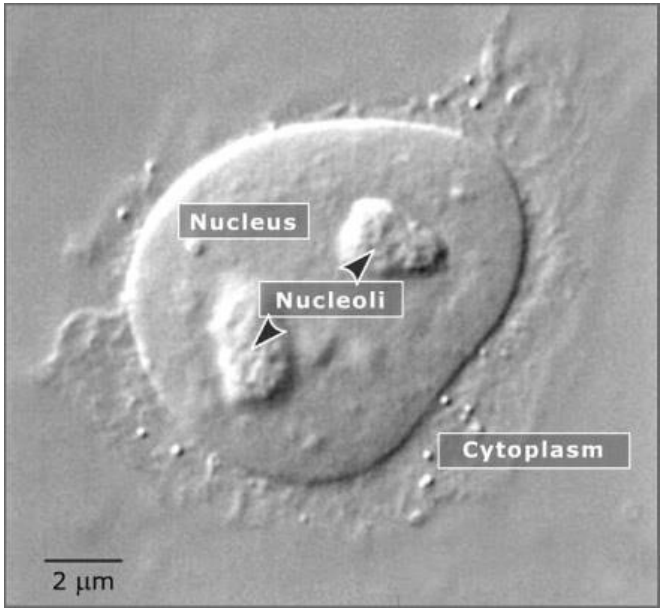


Figure: A cell from the human cervical carcinoma cell line HeLa has a nucleus that is easily seen using light microscopy.

Major compartments

1. Nuclear envelope
2. Nucleoplasm
3. Nucleolus
4. Nuclear Lamina
5. Chromatin

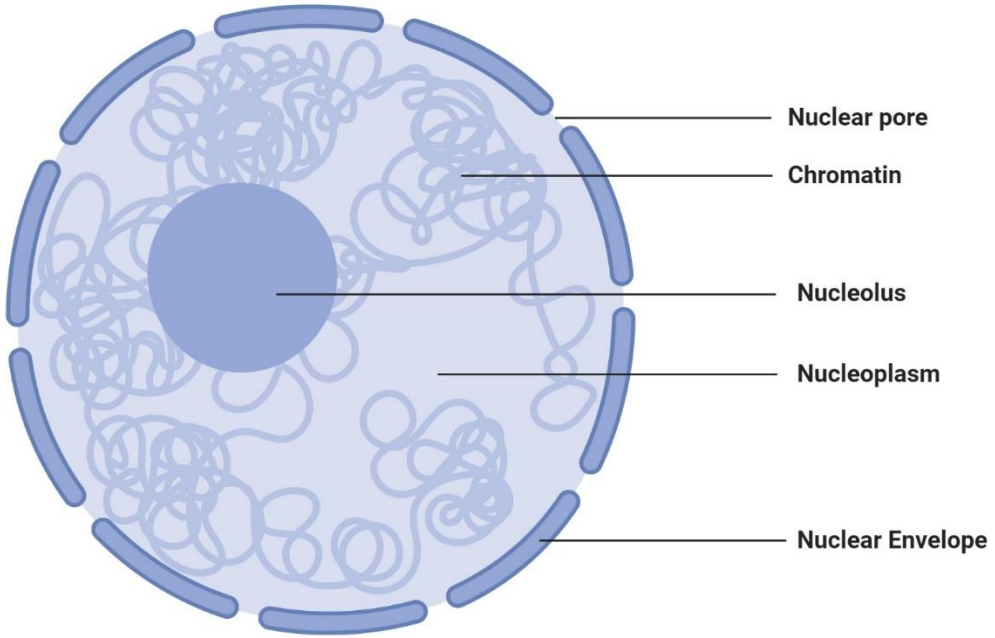


Fig. Nucleus, image copyright Sagar Aryal

1. The nuclear envelope

-Double membranes

- The outer nuclear membrane is continuous with the membranes of the ER, and the lumen of the nuclear envelope is continuous with the lumen of the ER.

-The nuclear envelope contains numerous NPCs

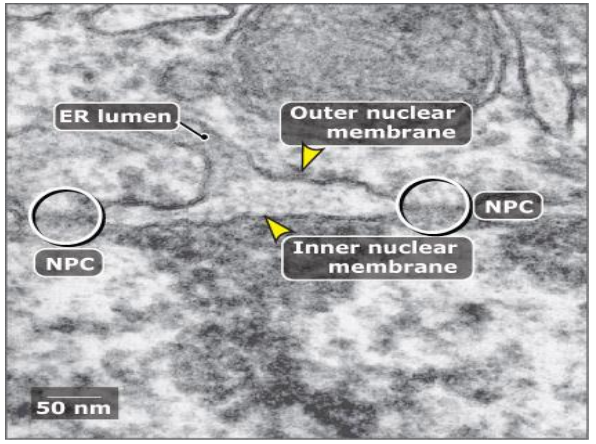


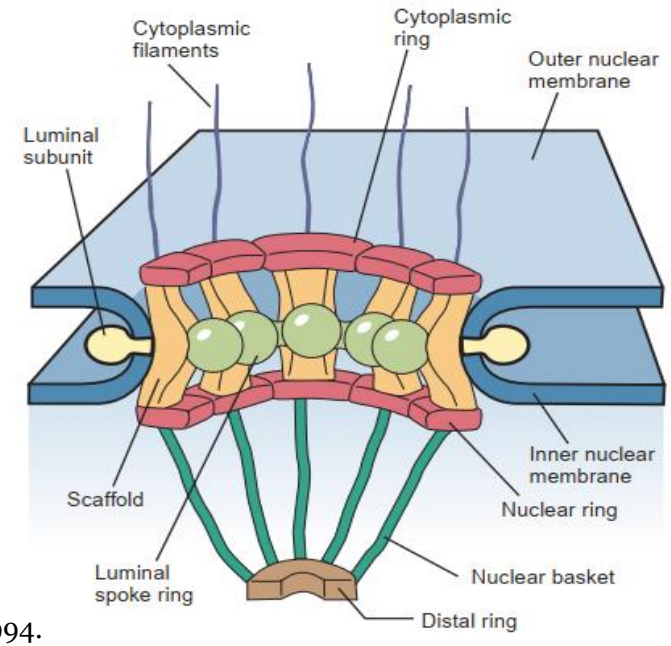
Figure 09.F12: The nuclear envelope is continuous with the endoplasmic reticulum.

1. The nuclear envelope - NPCs

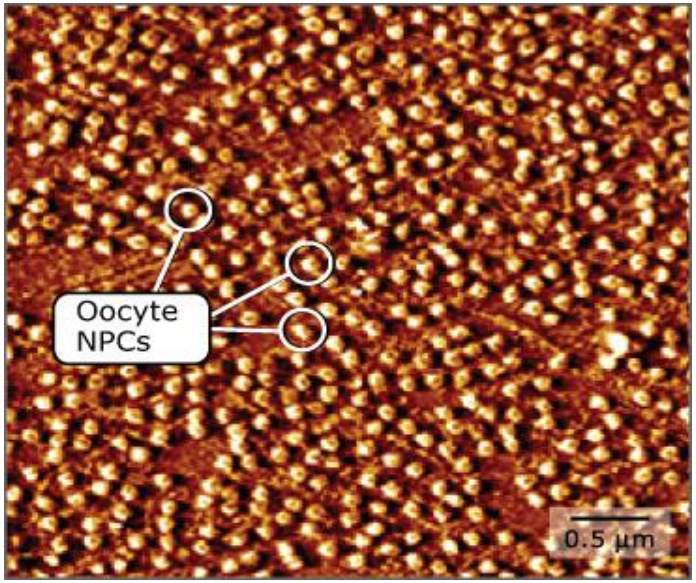
Large molecules are actively transported between the nucleus and cytoplasm

Uncharged molecules < 100 Da: pass through the nuclear envelope.
> 100 Da: NPCs

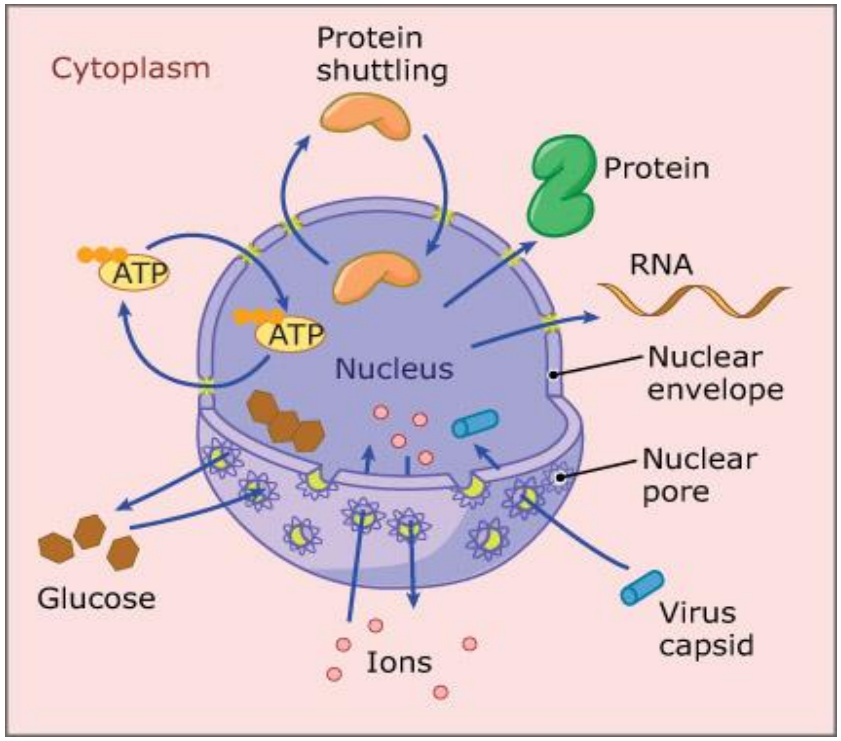
Particles up to 9 nm in diameter (corresponding to globular proteins up to 40 kDa) can pass through NPCs by passive diffusion, as can metabolites, nucleotides, and other small molecules



The nuclear pore complex.
(Modified with permission from Alberts B, Bray D, Lewis J, et al.: Molecular Biology of the Cell, 3rd ed. New York, Garland Publishing, 1994.



The nuclear pore is made of about 30-100 nucleoporin proteins.



Many different classes of molecules and macromolecules are transported through NPCs. Not shown are small, uncharged molecules (<100 Da) that can diffuse through the membranes of the nuclear envelope

Proteins are selectively transported into the nucleus through NPCs

Cores do not contain information for nuclear entry

→nuclear pore is a selective channel that allows only proteins with the proper information in their amino acid sequence to enter

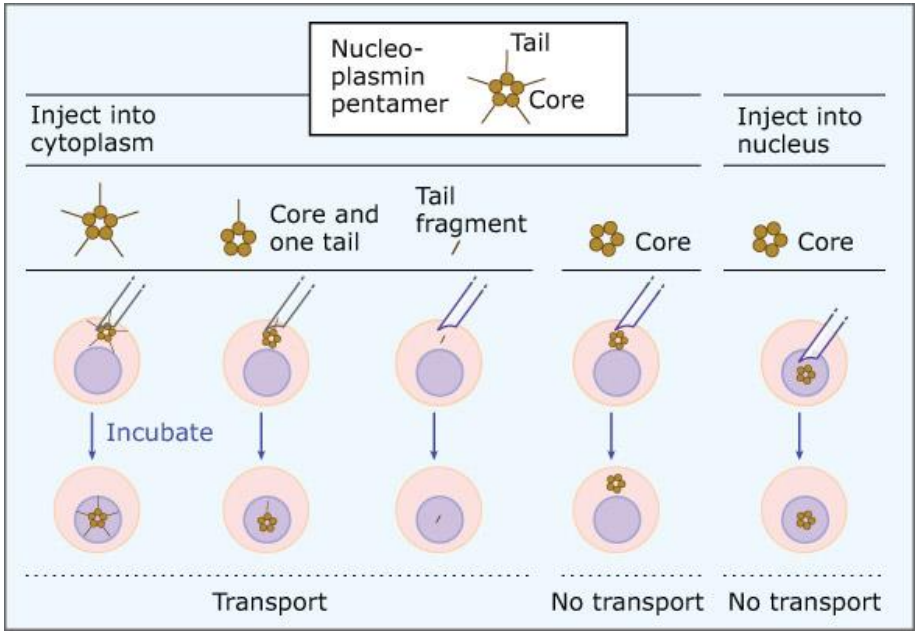


Figure: Injection of nucleoplasmin into the cytosol or the nucleus of *Xenopus oocytes* shows the importance of the C-terminal tail fragment.

❖Mature nuclear proteins contain sequence information required for their nuclear localization.

❖The signal within a protein that targets it to the nucleus is a stretch of amino acids termed the **NLS** (**nuclear localization sequence**)

Export of proteins from the nucleus is also receptor-mediated

Short stretches of amino acids rich in leucine act as the most common **nuclear export sequences (NESs)**

A nuclear export receptor binds proteins that contain **NESs** in the nucleus and transports them to the cytoplasm

The NESs were discovered in the course of studies of the growth of human HIV in infected cells

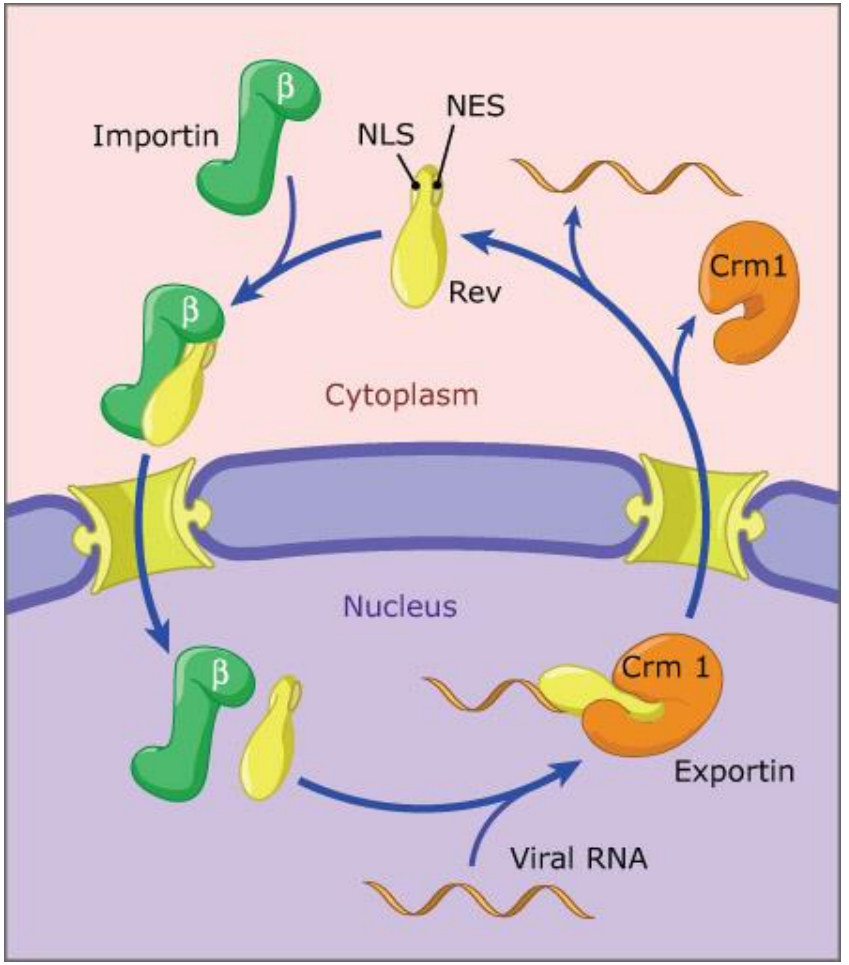
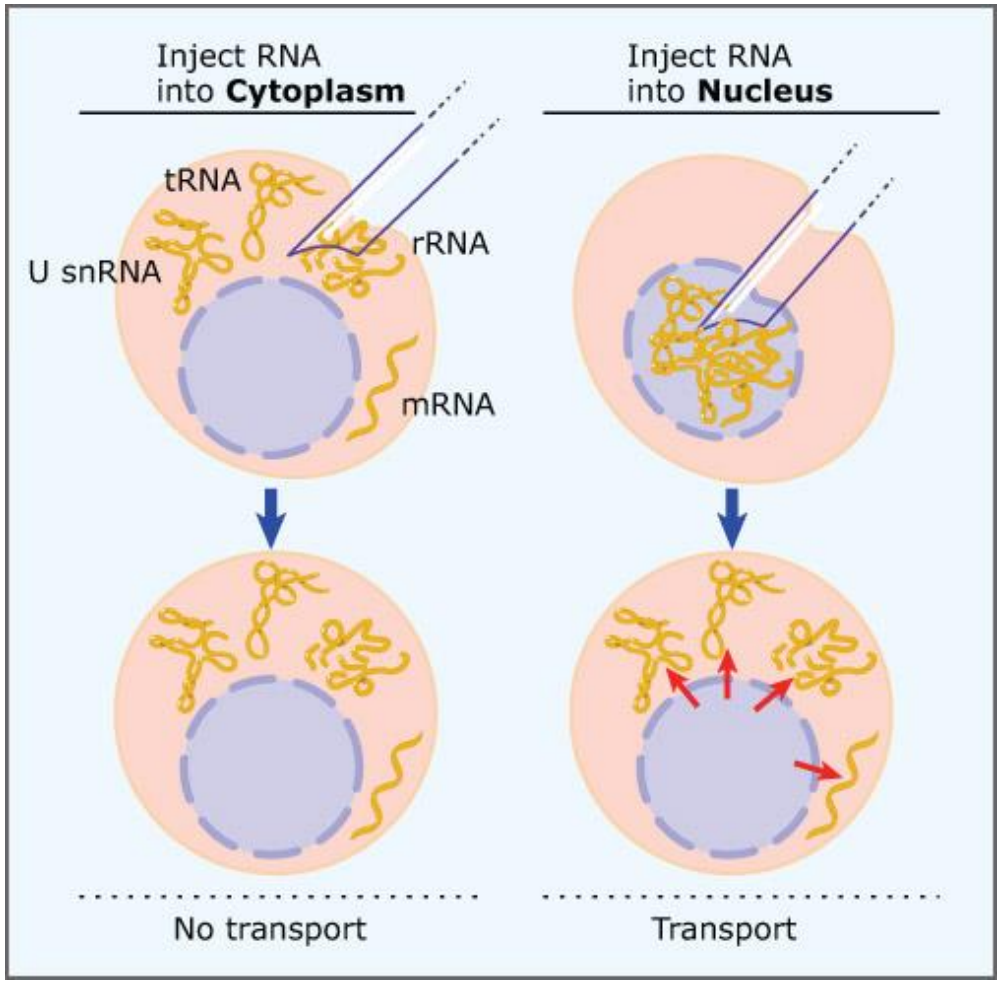


Figure: Some proteins, such as HIV Rev, shuttle between the nucleus and the cytoplasm.



Transport of most RNAs is **unidirectiona**l from the nucleus to the cytoplasm

2. The Nucleoplasm

- gelatinous substance within the nuclear envelope

- composed mainly of water with dissolved salts, enzymes, and organic molecules

- supports the nucleus by helping to maintain its shape

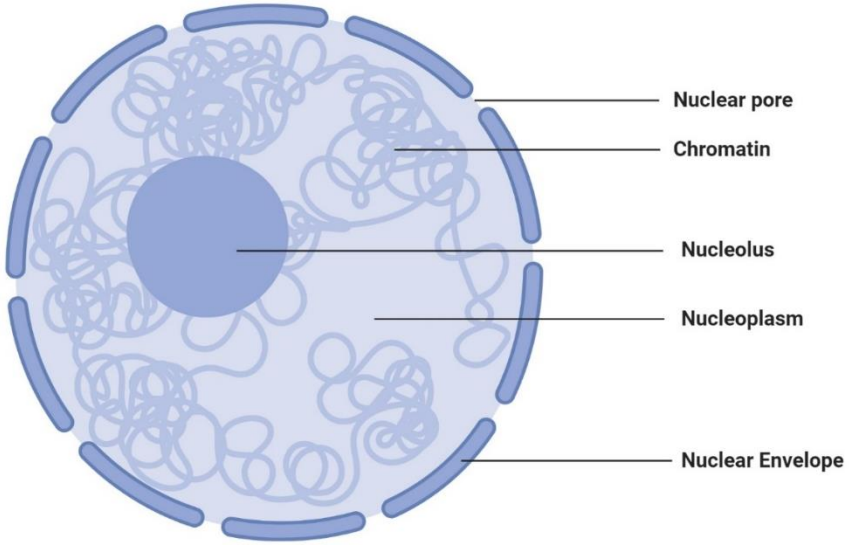


Fig. Nucleus, image copyright Sagar Aryal

3. The Nucleolus

- a dense, membrane-less structure within the nucleus

- Genes that encode rRNAs are present on multiple chromosomes that cluster together to form nucleolar subcompartments.

- rRNA is synthesized and ribosomal subunits are assembled in the **nucleolus.**

- disappears when a cell undergoes division and is reformed after the completion of cell division

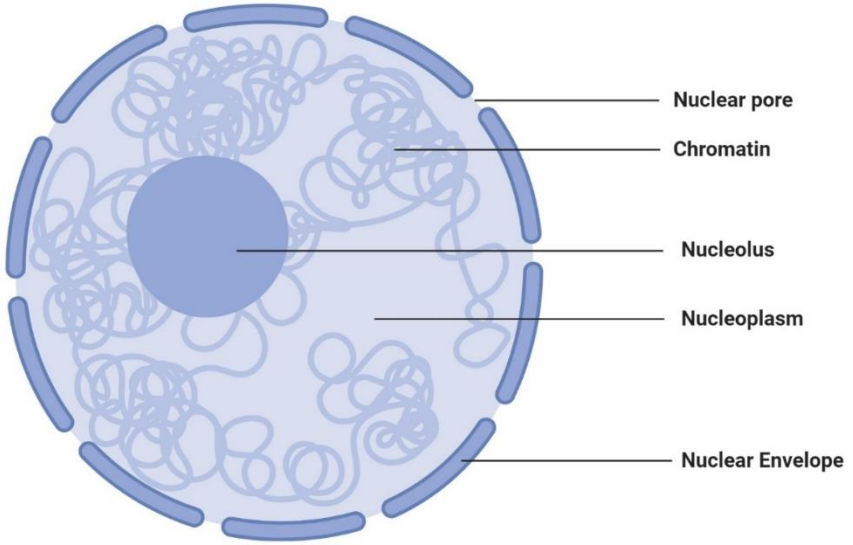


Fig. Nucleus, image copyright Sagar Aryal

4. The Nuclear Lamina

❖ Located **beneath the inner nuclear membrane** and is physically connected to it by lamina-associated integral membrane proteins.

❖ Plays a role in nuclear **envelope assembly** and may provide **physical support** for the nuclear envelope

❖ Protein complexes that interact with the nuclear lamina cross the nuclear envelope and link the cytoskeleton to the nuclear interior

❖Yeast and some other unicellular eukaryotes lack a nuclear lamina

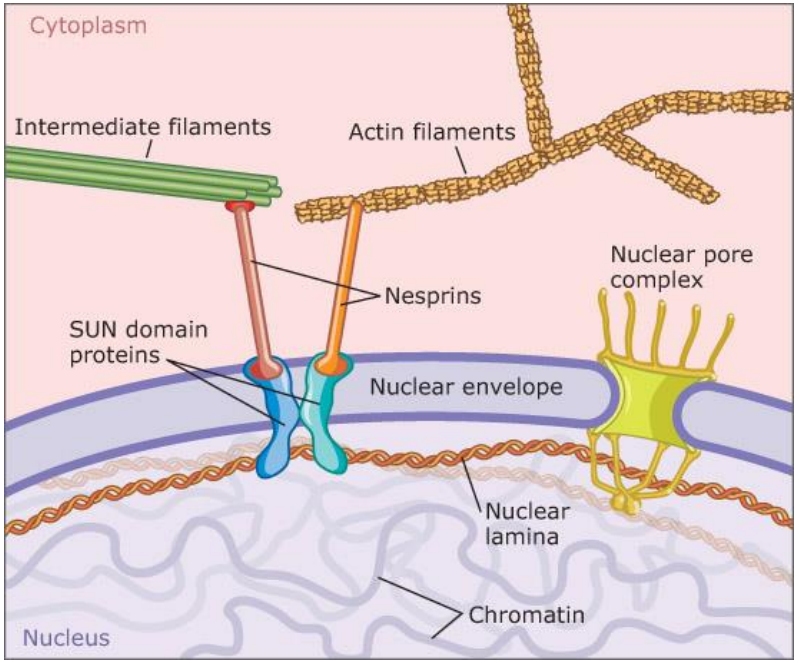
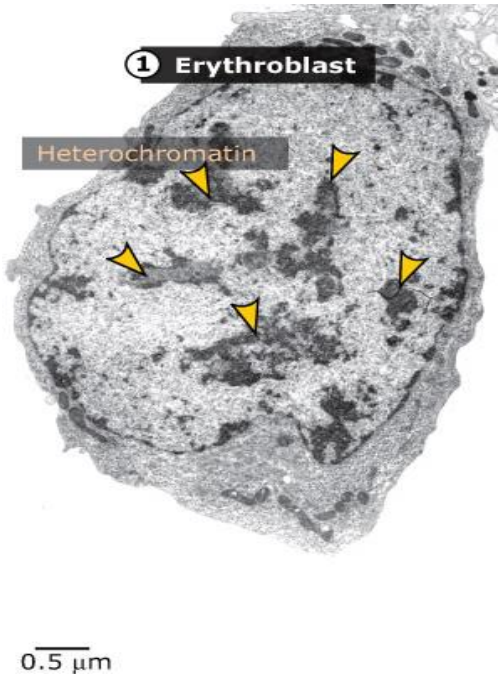


Figure " The nuclear lamina is anchored to the inner nuclear membrane

5. The Chromatin

- consist of DNA, which contains heredity information and instructions for cell growth, development, and reproduction

  - **Heterochromatin:** highly condensed, transcriptionally inactive form
  - **Euchromatin:** less densely compacted



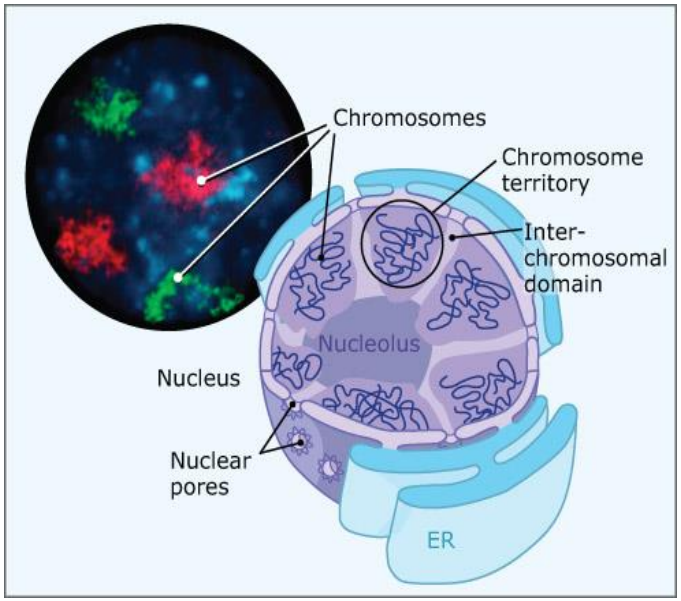


Figure: Individual chromosomes occupy distinct areas of the nucleus called chromosome territories.

What is the advantage to a eukaryotic cell of having a nucleus?

Having a nucleus allows a cell to have much more sophisticated regulation of gene expression than is possible in prokaryotic cells

The nucleus protects the DNA of the cell

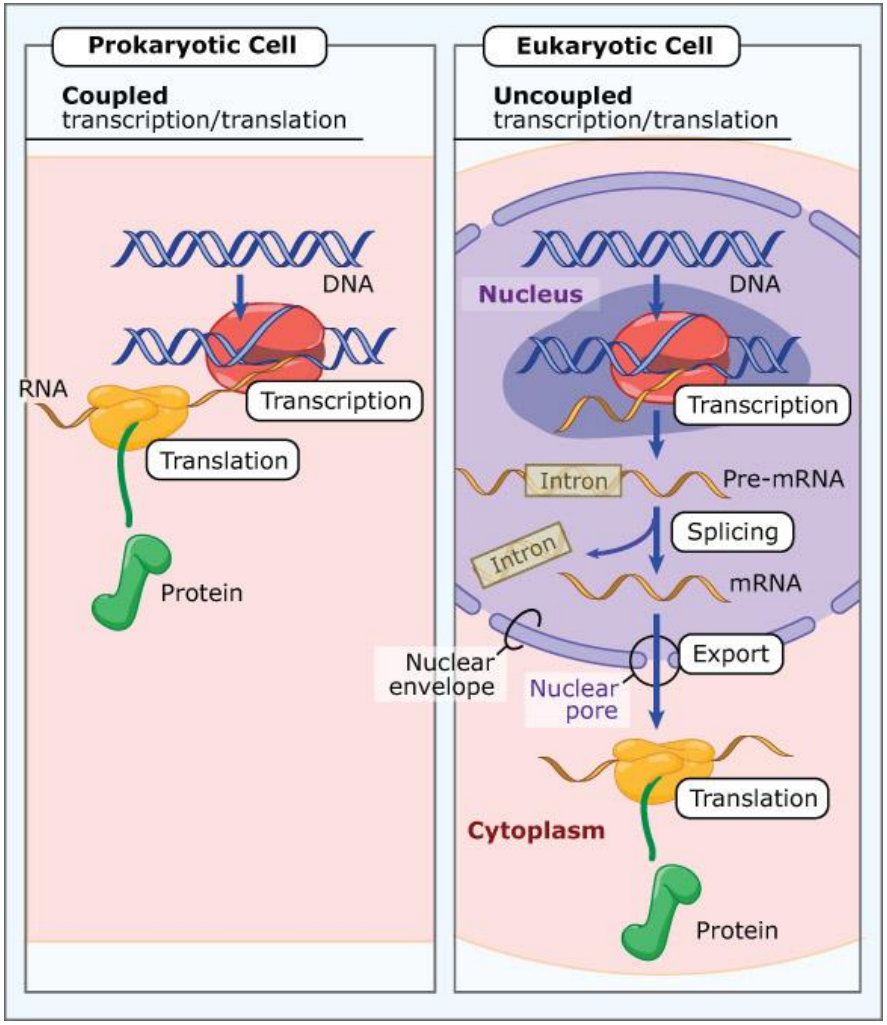


Figure: In prokaryotes, transcription and translation are coupled (left). In eukaryotes, transcription and translation occur in separate compartments (right)

Nuclei vary in appearance according to cell type and organism

- Nuclei range in size from about one micron (1 μm) to more than 10 μm in diameter

- Most cells have a single nucleus, but some cells contain multiple nuclei, and a few cell types lack nuclei

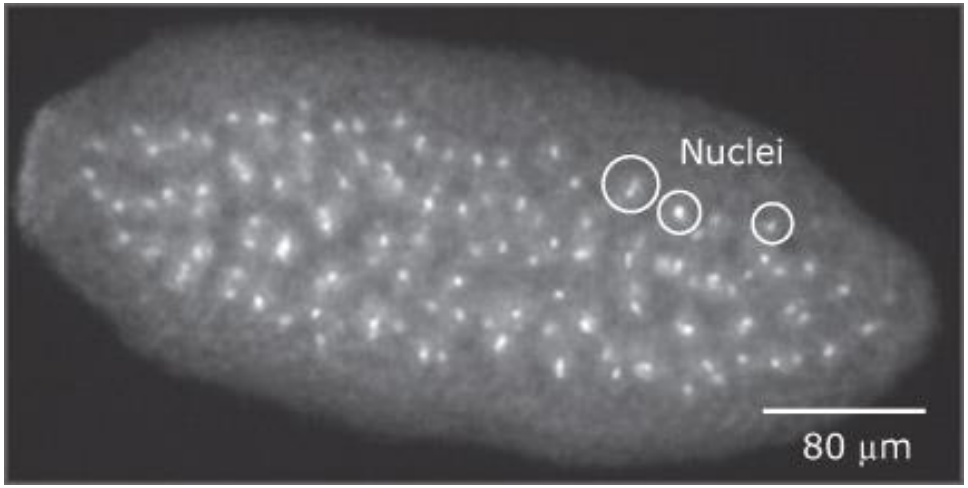
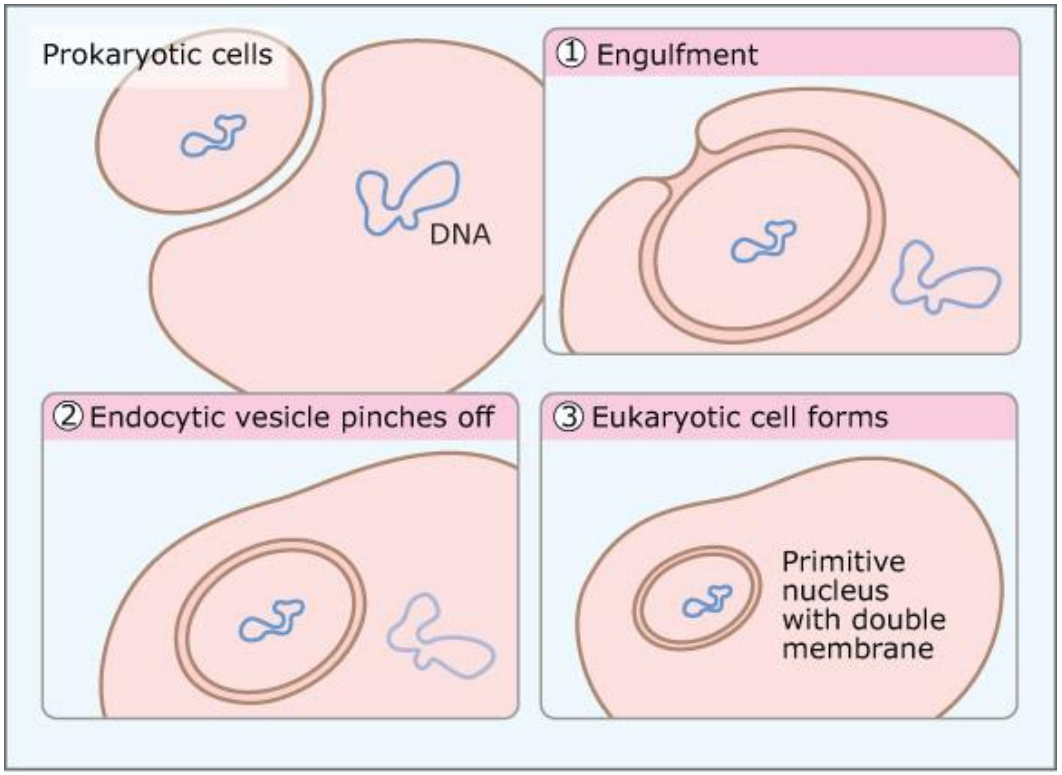
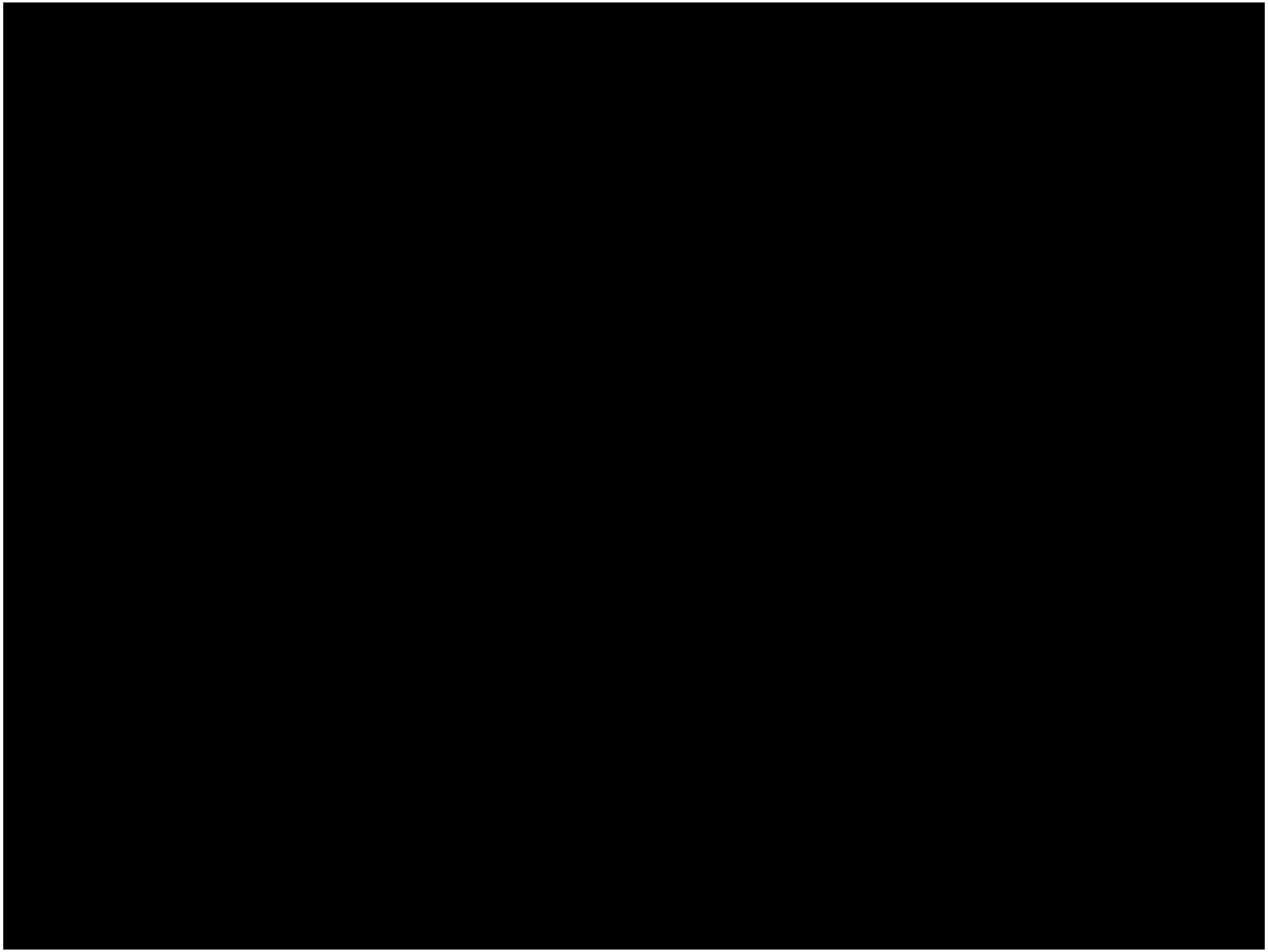


Figure: A *Drosophila* embryo at the multinucleate stage

Nucleus origin

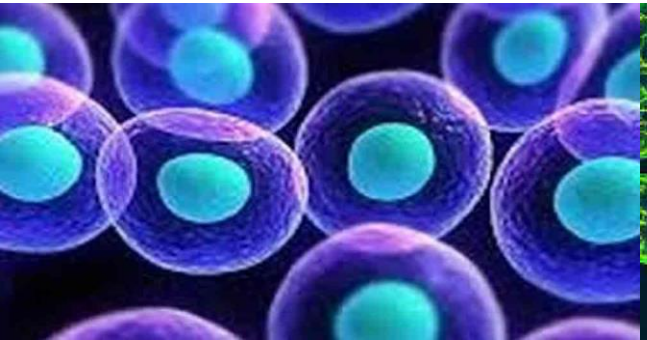


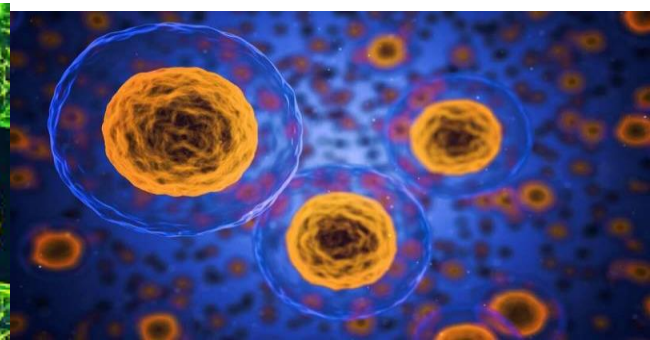
The nucleus may have arisen by **endosymbiosis**, a process in which one prokaryotic cell engulfs another cell, which then becomes a primitive nucleus



5.2

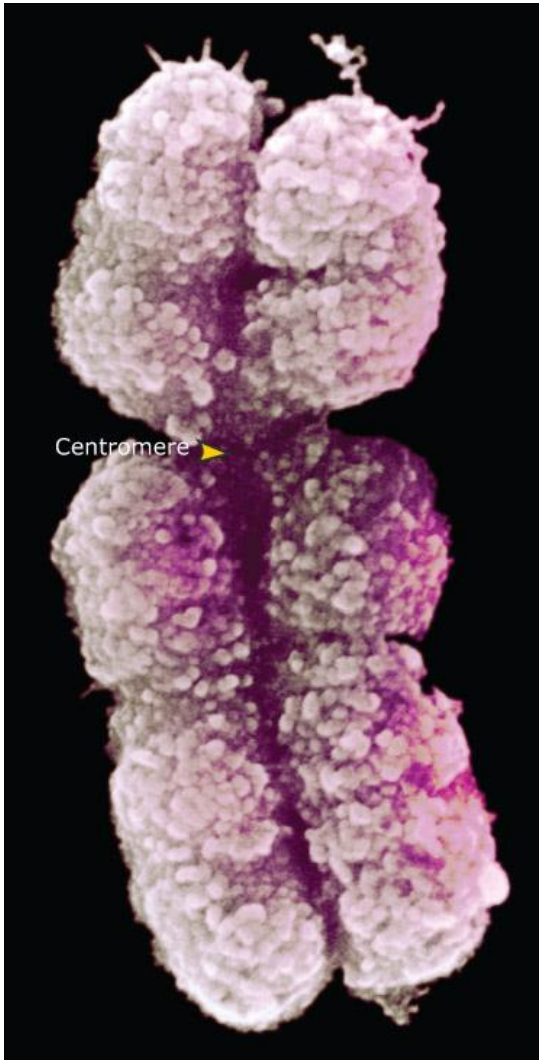
Chromatin – chromosomes





Bacteria- genetic material: in the form of a **nucleoid**

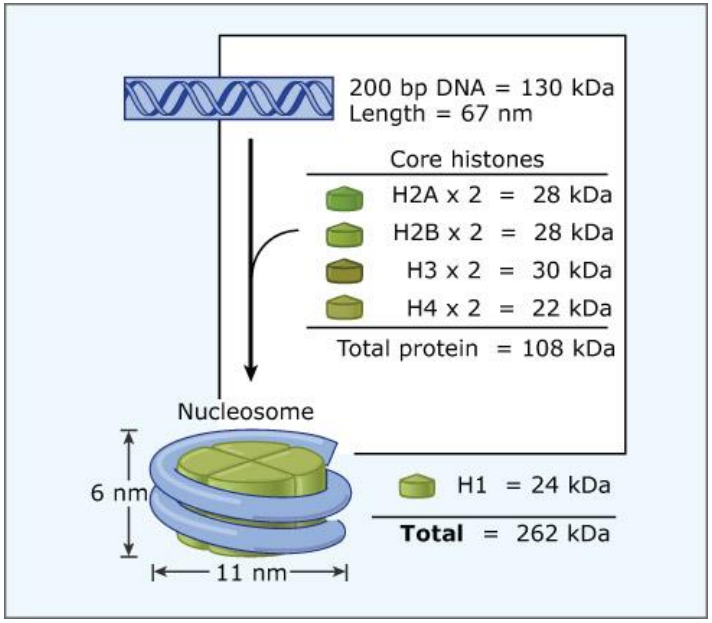
Eukaryotic- genetic material: the mass of **chromatin** within the nucleus



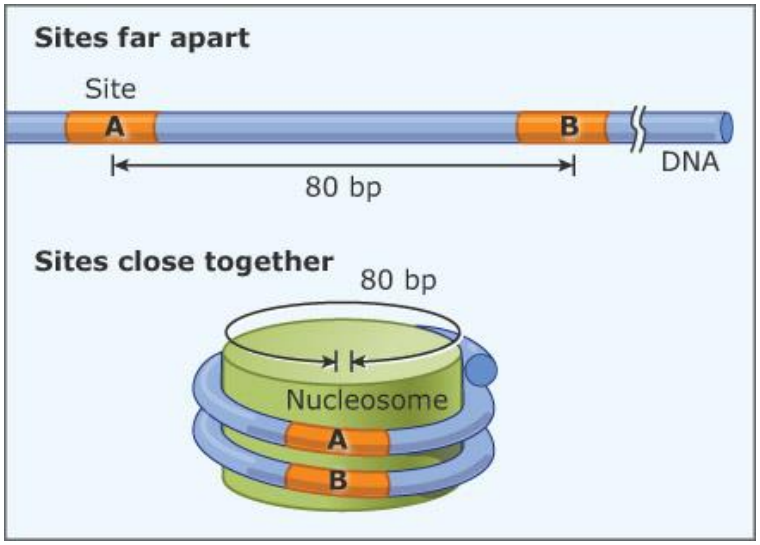
The **centromere** is the specialized DNA sequence of a chromosome that links a pair of sister chromatids

The nucleosome is the subunit of all chromatin

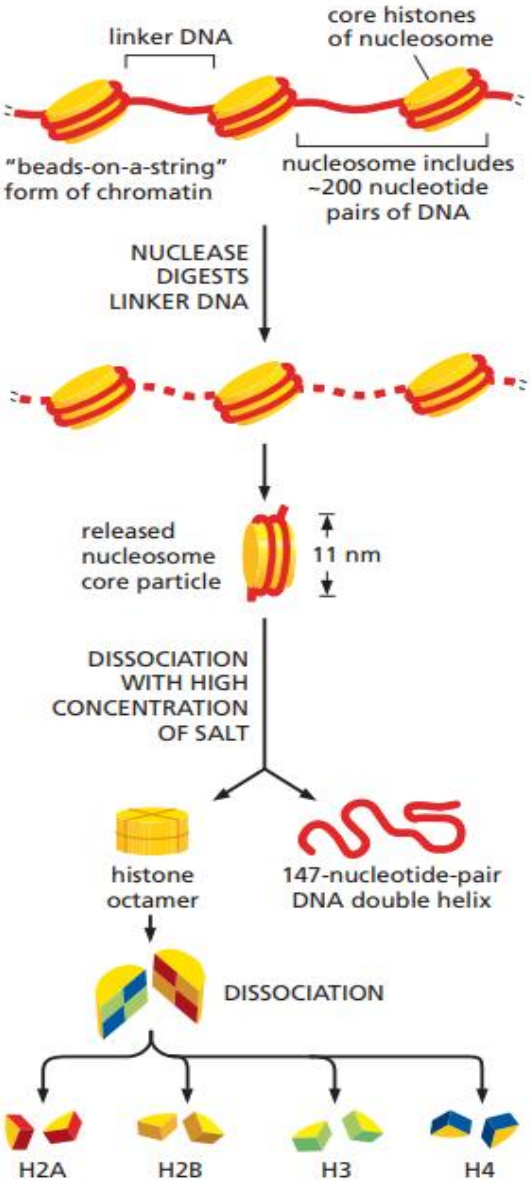
- Micrococcal nuclease releases individual nucleosomes from chromatin as ~10-nm particles.
- A nucleosome contains ~200 bp of DNA, two copies of each core histone (H2A, H2B, H3, and H4), and one copy of H1.
- DNA is wrapped around the outside surface of the protein octamer.

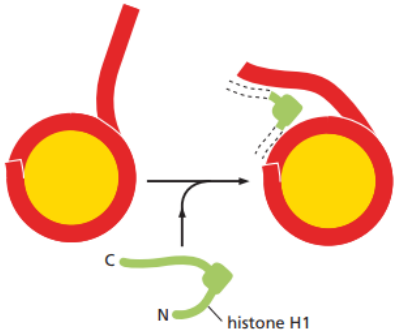


The nucleosome consists of approximately equal masses of DNA and histones..



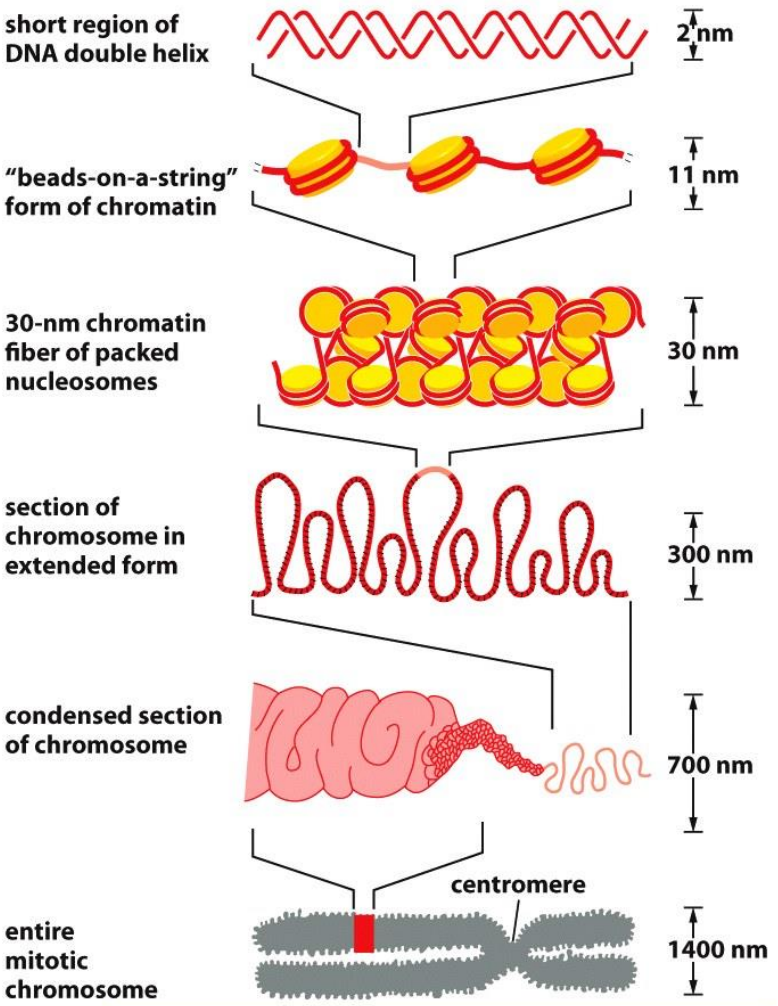
Sequences on the DNA that tie on different turns around the nucleosome may close together.

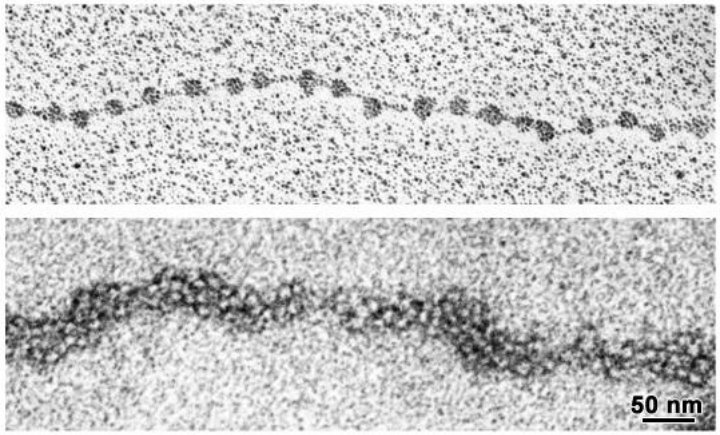




A linker histone helps to pull nucleosomes together and pack them into a more compact chromatin fiber.

Different levels of DNA packing





The 30 nm solenoid structure fiber forms loops whose base is attached to scaffold proteins.
Decondensation of these loops allows the transcription of genes they encode.

Chromosomes have banding patterns

- Certain staining techniques cause the chromosomes to have the appearance of a series of striations called G-bands.
- The bands are lower in G • C content than the interbands.
- Genes are concentrated in the G • C-rich interbands.

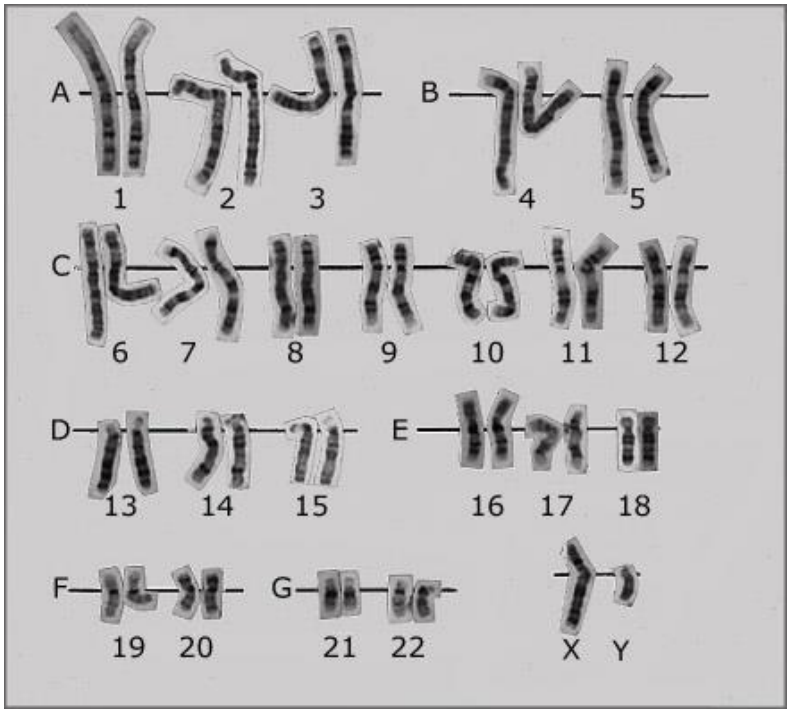
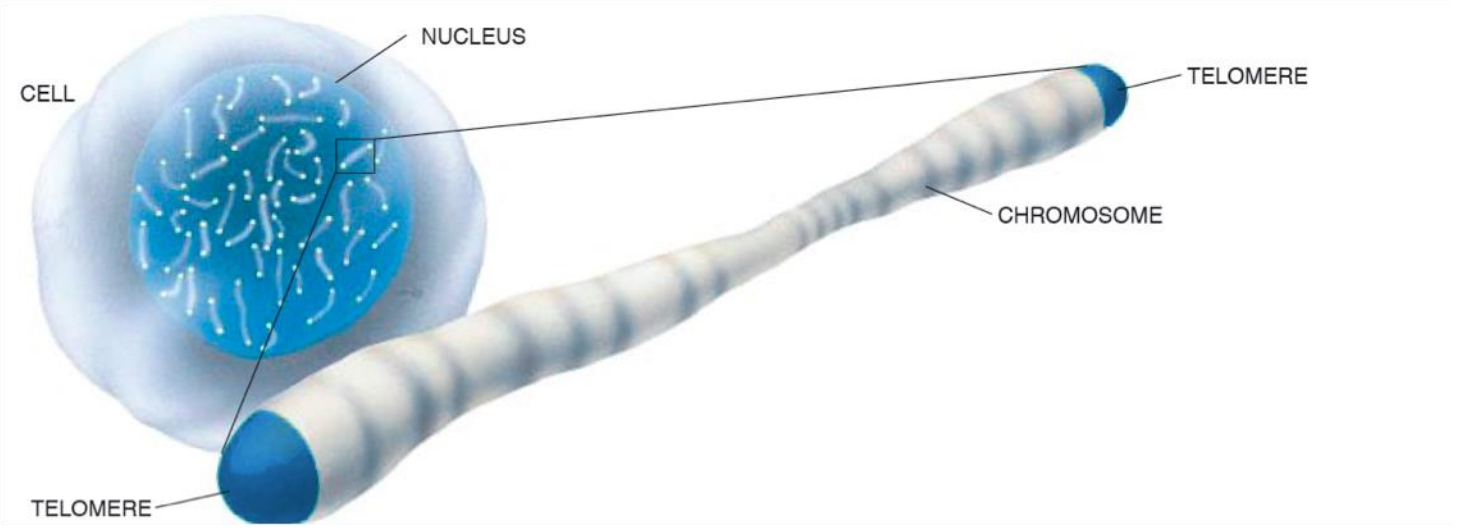


Figure: Every chromosome has a distinct G-banding pattern.

Telomeres are the ends of chromosomes



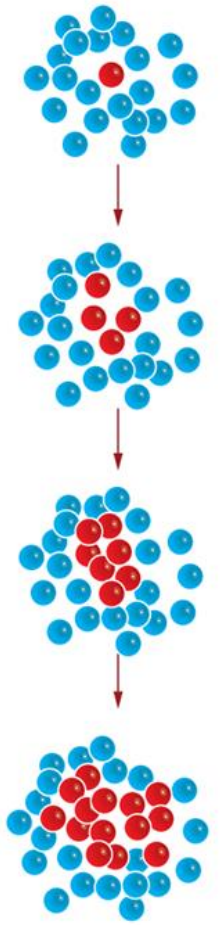
- **Protect ends**
- **Maintain length**

Telomeres are made of repeating sequences of TTAGGG on one strand paired with AATCCC on the other strand

Telomerase

It is required for cells that must divide many times

Cancer



The Nobel Prize in Physiology or Medicine 2009

"for the discovery of how chromosomes are protected by telomeres and the enzyme telomerase"



Photo: Gerbil, Licensed by Attribution Share Alike 3.0

Elizabeth H. Blackburn

1/3 of the prize

USA

University of California
San Francisco, CA, USA

b. 1948
(in Hobart, Tasmania, Australia)



Photo: Gerbil, Licensed by Attribution Share Alike 3.0

Carol W. Greider

1/3 of the prize

USA

Johns Hopkins University
School of Medicine
Baltimore, MD, USA

b. 1961



Photo © Harvard Medical School

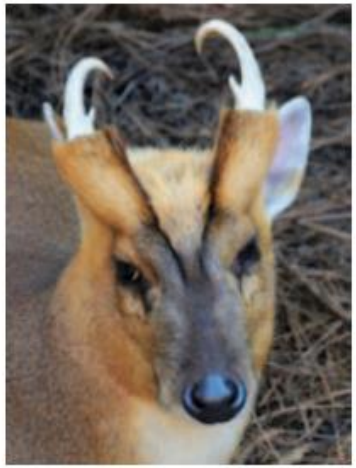
Jack W. Szostak

1/3 of the prize

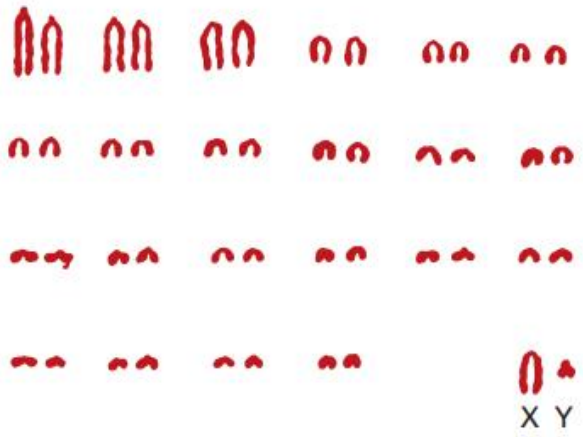
USA

Harvard Medical School;
Massachusetts General Hospital
Boston, MA, USA; Howard Hughes Medical Institute

b. 1952
(in London, United Kingdom)

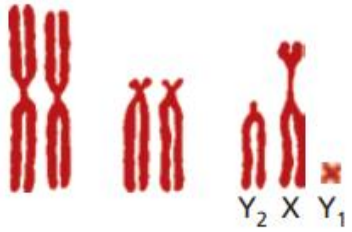


Chinese muntjac





Indian muntjac



Two closely related species can have similar genome sizes but very different chromosome numbers

The 23 chromosome pairs of the human genome



(A) (B)

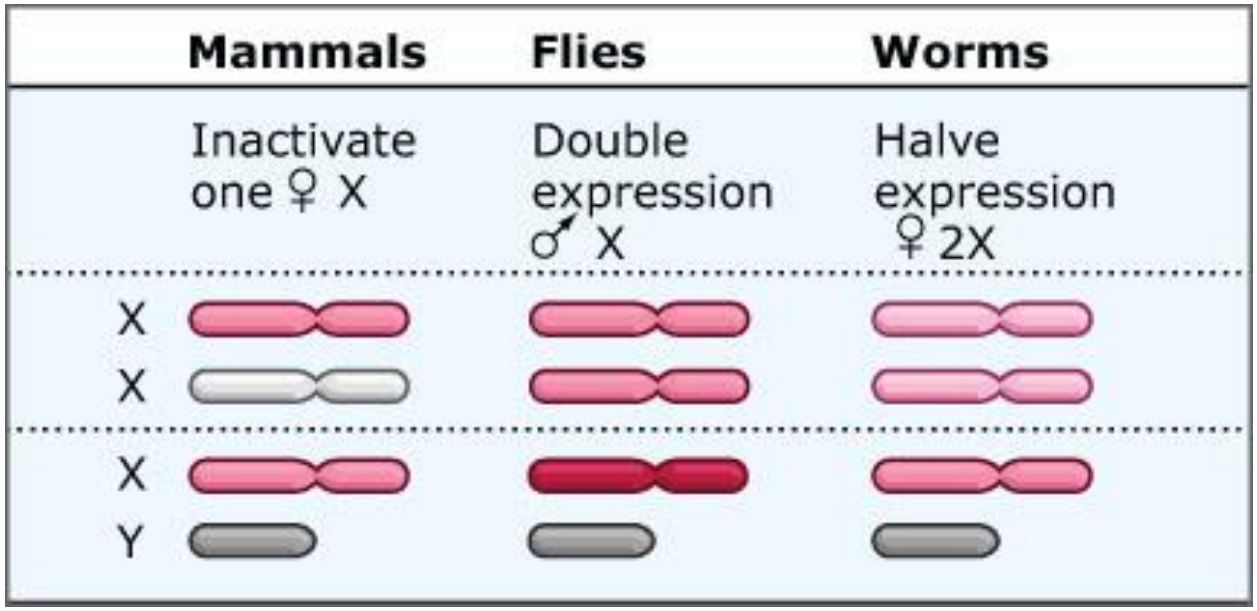
FISH: fluorescent in situ hybridization

Karyotype: chromosomes artificially lined up in order (e.g. for cytogenetic analysis)

X chromosomes undergo global changes

One of the two X chromosomes is inactivated at random in each cell during embryogenesis of eutherian mammals

Dosage compensation: equalizes the level of expression of X-linked genes in the two sexes



Different means of dosage compensation are used to equalize X chromosome expression in male and female.

One of the two X chromosomes is inactivated in the cells of mammalian females by heterochromatin formation. Each female cell contains two X chromosomes, one from the mother (Xm) and the other from the father (Xp).

